

Completions of Normed Vector Spaces

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Definition

A completion of a metric space (X, d) consists of:

- (1) A complete metric space (Z, ρ) ;
- (2) An isometry $\iota : X \rightarrow Z$ such that $\overline{\iota(X)}^\rho = Z$.

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Lemma

Let (X, d) be a metric space, (Y, ρ) a complete metric space, and $\iota : X \rightarrow Y$ an isometry. Then $(\overline{\iota(X)}^\rho, \rho, \iota)$ is a completion of (X, d) .

Theorem

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Sketch of Proof.

Let (Y, ρ, ι) and (Z, σ, j) be completions of (X, d) .

Let $y \in Y$. Since $\iota(X)$ is dense in Y , there exists a sequence $\{\iota(x_n)\}_n \subset \iota(X)$ such that $\iota(x_n) \rightarrow y$.

Since the sequence $\{\iota(x_n)\}_n$ is convergent, it is ρ -Cauchy.

Since ρ and σ are isometries, we have:

$$\sigma(j(x_n), j(x_m)) = d(x_n, x_m) = \rho(\iota(x_n), \iota(x_m)).$$

Hence if $\{\iota(x_n)\}_n$ is ρ -Cauchy, then $\{j(x_n)\}_n$ is σ -Cauchy.

Since (Z, σ, j) is complete, $\{j(x_n)\}_n$ is convergent.

Sketch of Proof (continued).

Define $\varphi : Y \rightarrow Z$ by $\varphi(y) = \lim_{n \rightarrow \infty} j(x_n)$.

To show uniqueness, show that φ is well-defined, an isometry, and bijective.

To show existence: Fix an $x_0 \in X$ and let $x \in X$ be arbitrary. Define $f_x : X \rightarrow \mathbb{R}$ by $f_x(t) = d(t, x) - d(t, x_0)$. Show that $f_x \in C_b(X)$ and that $\iota : X \rightarrow C_b(X)$ given by $x \mapsto f_x$ is an isometry. □

Question

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Remarks

- (1) Any normed vector space $(V, \|\cdot\|)$ induces a metric by the formula $d(x, y) := \|y - x\|$.
- (2) In the previous proof, since ι was not linear, if X was a normed vector space then $\overline{\iota(X)}^p$ wouldn't necessarily be a vector space.
- (3) We want to find a linear isometry into a "natural" Banach space.

Definition

Let $(V, \|\cdot\|)$ and $(W, \|\cdot\|)$ be normed $\mathbb{F}$ -vector spaces.

(1) The operator norm of $T \in \text{Hom}_{\mathbb{F}}(V, W)$ is:

$$\|T\|_{\text{op}} := \sup\{\|T(v)\| \mid v \in B_V\}$$

(2) $T \in \text{Hom}_{\mathbb{F}}(V, W)$ is called bounded if $\|T\|_{\text{op}} < \infty$. The set of all bounded linear maps from V to W is denoted:

$$\mathbb{B}(V, W) := \{T \in \text{Hom}_{\mathbb{F}}(V, W) \mid \|T\|_{\text{op}} < \infty\}$$

(3) The dual space of V is $V^{\vee} := \text{Hom}_{\mathbb{F}}(V, \mathbb{F})$.

(4) The continuous dual space of V is $V^* := \mathbb{B}(V, \mathbb{F})$.

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(4) The continuous dual space of V is $V^* := \mathbb{B}(V, \mathbb{F})$.

Remarks

(1) For any $v \in V$, $\|T(v)\| \leq \|T\|_{\text{op}} \cdot \|v\|$.

(2) A linear map is continuous if and only if it is bounded.

(3) $(\mathbb{B}(V, W), \|\cdot\|_{\text{op}})$ is a Banach space.

(4) Is V^* nonempty?

Theorem (Hahn-Banach)

Let X be an $\mathbb{R}$ -vector space, M a subspace of X , and $p : X \rightarrow \mathbb{R}$ a seminorm on X . If $f : M \rightarrow \mathbb{R}$ is a linear functional on M such that $|f(m)| \leq p(m)$ for all $m \in M$, then there exists a linear functional $F : X \rightarrow \mathbb{R}$ such that $F|_M = f$ and $|F(x)| \leq p(x)$ for all $x \in X$.

Corollary

Let $(V, \|\cdot\|)$ be a normed $\mathbb{R}$ -vector space. For any nonzero $v \in V$, there exists $F \in V^$ such that $F(v) = \|v\|$.*

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Proof.

Fix $v \in V$ and define $M := \{\alpha v \mid \alpha \in \mathbb{R}\}$. Define $f : M \rightarrow \mathbb{R}$ by $f(\alpha v) = \alpha \|v\|$.

Let $m \in M$. Then $m = \alpha v$ for some $\alpha \in \mathbb{R}$. Note that
 $|f(m)| = |f(\alpha v)| = |\alpha| \|v\| = \|\alpha v\| = \|m\|$.

Since every norm is automatically a seminorm, by the Hahn-Banach Theorem f extends to a linear functional $F : V \rightarrow \mathbb{R}$ with $F|_M = f$ and $|F(x)| \leq \|x\|$ for all $x \in V$.

Hence:

$$F(v) = F\left(\|v\| \cdot \frac{v}{\|v\|}\right) = \|v\| F\left(\frac{v}{\|v\|}\right) = \|v\| f\left(\frac{v}{\|v\|}\right) = \frac{\|v\|}{\|v\|} \|v\| = \|v\|.$$

Since norms are finite by definition, $F \in V^*$. □

Corollary

Let $(V, \|\cdot\|)$ be a normed $\mathbb{R}$ -vector space and $v \in V$ nonzero. Then $\|v\| = \sup\{|\varphi(v)| \mid \varphi \in B_{V^*}\}$.

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Proof.

If $\varphi \in B_{V^*}$, then $|\varphi(v)| \leq \|\varphi\|_{\text{op}} \|v\| \leq \|v\|$.

Hence $\sup\{|\varphi(v)| \mid \varphi \in B_{V^*}\} \leq \|v\|$.

By the previous corollary, find $\varphi_v \in V^*$ such that $\varphi_v(v) = \|v\|$.

Hence $\sup\{|\varphi(v)| \mid \varphi \in B_{V^*}\} \geq |\varphi_v(v)| = \|v\|$. □

Theorem

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Proof.

Let $v \in V$. Define $\widehat{v} : V^* \rightarrow \mathbb{R}$ by $\widehat{v}(\varphi) = \varphi(v)$. Easy to verify that $\widehat{v} \in (V^*)^\vee$.

By the previous corollary:

$$\begin{aligned}\|\widehat{v}\|_{\text{op}} &= \sup\{|\widehat{v}(\varphi)| \mid \varphi \in B_{V^*}\} \\ &= \sup\{|\varphi(v)| \mid \varphi \in B_{V^*}\} \\ &= \|v\| \\ &< \infty.\end{aligned}$$

Hence $\widehat{v} \in V^{**}$. Define $\iota : V \rightarrow V^{**}$ by $v \mapsto \widehat{v}$. This map is an isometry since $\|\iota(v)\|_{\text{op}} = \|\widehat{v}\|_{\text{op}} = \|v\|$.

Proof (continued).

Let $v_1, v_2 \in V$ and $\alpha \in \mathbb{R}$. We have:

$$\begin{aligned}\iota(v_1 + \alpha v_2)(\varphi) &= \widehat{v_1 + \alpha v_2}(\varphi) \\ &= \varphi(v_1 + \alpha v_2) \\ &= \varphi(v_1) + \alpha \varphi(v_2) \\ &= \widehat{v_1}(\varphi) + \alpha \widehat{v_2}(\varphi) \\ &= (\iota(v_1) + \alpha \iota(v_2))(\varphi).\end{aligned}$$

Thus ι is linear. □

Remarks

- (1) Using the same terminology from the previous theorem, the completion of a normed $\mathbb{R}$ -vector space $(V, \|\cdot\|)$ is $(\overline{\iota(V)}, \|\cdot\|_{\text{op}}, \iota)$.
- (2) The completion of $(V, \|\cdot\|)$ is unique up to isometric isomorphism.
- (3) The closure of a linear subspace is also a linear subspace.
- (4) If V is finite-dimensional, then $\iota(V) \cong V^{**}$. If V were infinite-dimensional, then this may or may not be the case.
- (5) V is called reflexive if $\iota(V) \cong V^{**}$.

Thank you!